



Reference Image



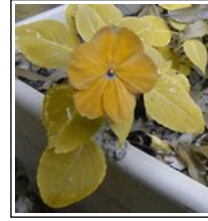
Deuteranomaly (4 nm)



Deuteranomaly (8 nm)



Deuteranomaly (12 nm)



Deuteranomaly (16 nm)



Deuteranopia

A Physiologically-based Model for Simulation of Color Vision Deficiency

Gustavo M. Machado
gmmachado@inf.ufrgs.br

Manuel M. Oliveira
oliveira@inf.ufrgs.br

Leandro A. F. Fernandes
laffernandes@inf.ufrgs.br

Instituto de Informática, UFRGS



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Abstract

Color vision deficiency (CVD) affects approximately 200 million people worldwide, compromising the ability of these individuals to effectively perform color and visualization-related tasks. This has a significant impact on their private and professional lives. We present a physiologically-based model for simulating color perception. Our model is based on the stage theory of human color vision and is derived from data reported in electrophysiological studies. It is the first model to consistently handle normal color vision, anomalous trichromacy, and dichromacy in a unified way. We have validated the proposed model through an experimental evaluation involving groups of color vision deficient individuals and normal color vision ones. Our model can provide insights and feedback on how to improve visualization experiences for individuals with CVD. It also provides a framework for testing hypotheses about some aspects of the retinal photoreceptors in color vision deficient individuals.

Downloads



[Full paper \(with correct Eqs. \(17\) and \(18\)\)](#)

Errata

There is a typo in Eqs (17) e (18) of our Vis 2009 paper (page 1295 of the conference proceedings). Factors that should be multiplying the last term inside parenthesis, happen to appear outside the parenthesis. We would like to point out that the images shown in the paper and the matrices provided as supplementary material are correct (they were computed using the correct version of Eqs. 17 e 18). The error happened during the typing of the paper. The correct equations can be seen below:

$$L_{\alpha}(\lambda) = \alpha L(\lambda) + 0.96 \frac{Area_L}{Area_M} (1 - \alpha) M(\lambda) \quad (17)$$

$$M_{\alpha}(\lambda) = \alpha M(\lambda) + \frac{1}{0.96} \frac{Area_M}{Area_L} (1 - \alpha) L(\lambda) \quad (18)$$

Results

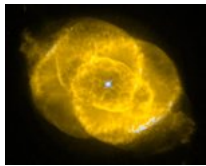
Video



06:43

Images

The images below illustrate examples of how colors can be confused for individuals with CVD. A set of scientific visualization and information visualization images simulating for normal trichromats the perception of protanomalous and deuteranomalous with severities of 2 nm, 8 nm, 14 nm, and the perception of dichromats which is equivalent to anomalous trichromacy with severity of 20 nm.

Type	Normal color vision	2 nm	8 nm	14 nm	Dichromacy
Prot					
Deut					
Prot					
Deut					
Prot					
Deut					

Tutorial

Individuals with color vision deficiency (CVD) represents approximately 200 million of people worldwide. It has, in most of the cases, genetic cause and there is no cure or treatment for those individuals. Thus, understanding how is color perceived by those individuals is of high relevance. In this tutorial we explain how to use this model to simulate the perception of individuals with CVD.

According to this model the simulation of the perception of individuals with CVD is given by a single matrix multiplication $\mathcal{M}_{CVD}$ as in Eq. 1 below. The RGB vector on the right represents the reference RGB color, which is multiplied by the matrix $\mathcal{M}_{CVD}$ to compute the simulated color $R_S G_S B_S$.

(1)

This matrix can be computed for all severities of protanomaly, deuteranomaly, and tritanomaly. Aiming efficiency, one can pre-compute the matrices for many severities and use them in the application allowing to switching between many severities and types of CVDs. Table 1 contains a set of pre-computed matrices for severities in the range $[0.0, 1.0]$ where 1.0 represents the highest severity or a case of dichromacy, and 0.0 represents absence of CVD.

The matrices in Table 1 were computed for severities growing with a step of 0.1. If you want to simulate colors with higher precision severities values, for example, a severity of 0.873, you can use the model in the article to compute the specified matrix. This is the most accurate approach, but it is also possible to interpolate between the two matrices with nearest severities. For example, to compute the matrix for a case of severity 0.873, the matrices 0.8 and 0.9 can be interpolated with a weight of 0.73. This approach is a fast approximation and also very accurate for a set of pre-computed matrices with severity step of 0.1.

Table 1: Simulation matrices $\mathcal{M}_{CVD}$

Severity	Protanomaly			Deuteranomaly			Tritanomaly		
0.0	1.000000	0.000000	-0.000000	1.000000	0.000000	-0.000000	1.000000	0.000000	-0.000000
	0.000000	1.000000	0.000000	0.000000	1.000000	0.000000	0.000000	1.000000	0.000000
	-0.000000	-0.000000	1.000000	-0.000000	-0.000000	1.000000	-0.000000	-0.000000	1.000000
0.1	0.856167	0.182038	-0.038205	0.866435	0.177704	-0.044139	0.926670	0.092514	-0.019184
	0.029342	0.955115	0.015544	0.049567	0.939063	0.011370	0.021191	0.964503	0.014306
	-0.002880	-0.001563	1.004443	-0.003453	0.007233	0.996220	0.008437	0.054813	0.936750
0.2	0.734766	0.334872	-0.069637	0.760729	0.319078	-0.079807	0.895720	0.133330	-0.029050
	0.051840	0.919198	0.028963	0.090568	0.889315	0.020117	0.029997	0.945400	0.024603
	-0.004928	-0.004209	1.009137	-0.006027	0.013325	0.992702	0.013027	0.104707	0.882266
0.3	0.630323	0.465641	-0.095964	0.675425	0.433850	-0.109275	0.905871	0.127791	-0.033662
	0.069181	0.890046	0.040773	0.125303	0.847755	0.026942	0.026856	0.941251	0.031893
	-0.006308	-0.007724	1.014032	-0.007950	0.018572	0.989378	0.013410	0.148296	0.838294
0.4	0.539009	0.579343	-0.118352	0.605511	0.528560	-0.134071	0.948035	0.089490	-0.037526
	0.082546	0.866121	0.051332	0.155318	0.812366	0.032316	0.014364	0.946792	0.038844
	-0.007136	-0.011959	1.019095	-0.009376	0.023176	0.986200	0.010853	0.193991	0.795156
0.5	0.458064	0.679578	-0.137642	0.547494	0.607765	-0.155259	1.017277	0.027029	-0.044306
	0.092785	0.846313	0.060902	0.181692	0.781742	0.036566	-0.006113	0.958479	0.047634
	-0.007494	-0.016807	1.024301	-0.010410	0.027275	0.983136	0.006379	0.248708	0.744913
0.6	0.385450	0.769005	-0.154455	0.498864	0.674741	-0.173604	1.104996	-0.046633	-0.058363
	0.100526	0.829802	0.069673	0.205199	0.754872	0.039929	-0.032137	0.971635	0.060503
	-0.007442	-0.022190	1.029632	-0.011131	0.030969	0.980162	0.001336	0.317922	0.680742
0.7	0.319627	0.849633	-0.169261	0.457771	0.731899	-0.189670	1.193214	-0.109812	-0.083402
	0.106241	0.815969	0.077790	0.226409	0.731012	0.042579	-0.058496	0.979410	0.079086
	-0.007025	-0.028051	1.035076	-0.011595	0.034333	0.977261	-0.002346	0.403492	0.598854
0.8	0.259411	0.923008	-0.182420	0.422823	0.781057	-0.203881	1.257728	-0.139648	-0.118081
	0.110296	0.804340	0.085364	0.245752	0.709602	0.044646	-0.078003	0.975409	0.102594
	-0.006276	-0.034346	1.040622	-0.011843	0.037423	0.974421	-0.003316	0.501214	0.502102
0.9	0.203876	0.990338	-0.194214	0.392952	0.823610	-0.216562	1.278864	-0.125333	-0.153531
	0.112975	0.794542	0.092483	0.263559	0.690210	0.046232	-0.084748	0.957674	0.127074
	-0.005222	-0.041043	1.046265	-0.011910	0.040281	0.971630	-0.000989	0.601151	0.399838
1.0	0.152286	1.052583	-0.204868	0.367322	0.860646	-0.227968	1.255528	-0.076749	-0.178779
	0.114503	0.786281	0.099216	0.280085	0.672501	0.047413	-0.078411	0.930809	0.147602
	-0.003882	-0.048116	1.051998	-0.011820	0.042940	0.968881	0.004733	0.691367	0.303900

Reference

Citation

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BibTeX

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Keywords

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